

Analysis of European Research and Innovation Projects Using CORDIS Data

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Abstract—This project presents an exploratory analysis of European research and innovation projects funded under the Horizon 2020 and Horizon Europe programmes using publicly available CORDIS data. The dataset contains information about projects, participating organizations, countries, and sectors involved in European Union research activities. Since the raw dataset included duplicated records, inconsistencies, and multiple participations of the same organizations, a data cleaning and harmonisation process was performed to improve accuracy and reliability. The study applies Python-based data preprocessing and exploratory data analysis (EDA) techniques to investigate the distribution of projects across different countries and sectors. Various visualization methods were used to identify participation trends, collaboration patterns, and the overall structure of the dataset. The analysis provides meaningful insights into how European research funding is distributed and highlights the role of organizations and countries in collaborative innovation projects. The project demonstrates the importance of data cleaning, analysis, and visualization in extracting useful information from large-scale real-world datasets.

Keywords—CORDIS, Horizon 2020, Horizon Europe, Data Analysis, Exploratory Data Analysis (EDA), Data Cleaning, European Union Research

1. INTRODUCTION

1.1. Background

The dataset used in this project was developed using public data from CORDIS, the official database of the European Commission that provides information on research and innovation projects funded by the European Union. The data were collected from both Horizon 2020 and Horizon Europe programmes, with updates available up to January 2026. Indeed, the datasets have been provided by [SoBigData Research Infrastructure, Pisa, Italy](#)

Before analysis, the dataset underwent a cleaning and harmonisation process to improve data quality and consistency. This included removing duplicate organisations, correcting inconsistencies, and merging multiple participations of the same entity within individual projects. These preprocessing steps reduced the dataset size while maintaining its original structure, accuracy, and reliability, making it more suitable for research and analytical purposes.

European research and innovation programmes play a significant role in supporting scientific development, technological advancement, and international collaboration among countries and institutions. The large amount of data generated by these programmes creates opportunities for data-driven research and analysis. However, analysing such datasets requires appropriate preprocessing and visualization methods to obtain meaningful insights.

1.2. Problem Statement

Research and innovation projects funded under Horizon 2020 and Horizon Europe involve many countries and organizations, generating large and complex datasets. Analysing this data is challenging due to duplicated records, multiple participations, and inconsistencies. Without proper preprocessing, the analysis may produce inaccurate or misleading results.

This project aims to study how European research projects are distributed across sectors and countries using cleaned and harmonised CORDIS data. The analysis focuses on understanding participation trends, collaboration patterns, and the role of different organizations in European research activities.

1.3. Research Questions

The research questions addressed in this project are:

1. How do collaboration structures shape innovation ecosystems within European Horizon research programmes?
2. Which organisations act as central collaboration hubs in European research networks?
3. Which countries dominate project participation, funding allocation, and coordination activities in Horizon projects?
4. How do university–industry collaborations influence project funding and consortium structures?
5. How have collaboration patterns evolved from Horizon 2020 to Horizon Europe over time?

1.4. Objectives

The objective of this Datathon is to explore and analyse real-world European research and innovation data collected from the Horizon 2020 and Horizon Europe programmes through the CORDIS database. The project focuses on understanding the structure and characteristics of the dataset by applying data cleaning, preprocessing, and exploratory data analysis (EDA) techniques using Python tools and libraries such as Pandas, NumPy, and Matplotlib. The study aims to remove duplicate records, correct inconsistencies, and harmonise the data to improve its quality and reliability for analysis. In addition, the project seeks to identify patterns and trends related to project participation across different countries, organizations, and sectors. Through statistical analysis and data visualization, the research examines collaboration patterns, participation distribution, and the overall contribution of institutions involved in European research projects. The project also aims to develop practical knowledge and technical skills in data analytics, visualization, and big data processing using Jupyter Notebook and Python-based analytical methods.

2. DATASET DESCRIPTION AND PREPROCESSING

The dataset used in this project was obtained from the CORDIS database, which provides open access information about European Union funded research and innovation projects. The dataset contains information related to project participation, organizations, countries, and research sectors involved in the Horizon 2020 and Horizon Europe programmes.

Several variables were selected for analysis, including:

- Project identifiers
- Organization names
- Country information
- Programme details
- Participation records
- Research sectors

The dataset initially contained duplicated entries and inconsistencies due to multiple participations of the same organizations in different projects. Therefore, data cleaning and preprocessing were necessary before performing the analysis.

The preprocessing stage included:

- Removing duplicate records
- Handling missing values

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- Correcting inconsistent organization names
- Harmonising participation records
- Filtering unnecessary columns

Python libraries such as Pandas and NumPy were used extensively during preprocessing. These steps improved the quality, consistency, and reliability of the dataset, making it suitable for exploratory analysis and visualization.

3. METHODOLOGY

The project was implemented using Python in a Jupyter Notebook environment. Various data analysis and visualization libraries were used throughout the project.

3.1. Tools and Libraries Used

The following Python libraries were used:

- Pandas
- NumPy
- Matplotlib
- Seaborn

Pandas was mainly used for data cleaning, filtering, grouping, and aggregation operations. NumPy supported numerical computations, while Matplotlib and Seaborn were used for creating visualizations and charts.

3.2. Data Preprocessing

The preprocessing stage involved several important steps:

1. Loading the dataset into Jupyter Notebook
2. Inspecting dataset structure and variables
3. Identifying missing and duplicated values
4. Cleaning inconsistent records
5. Merging repeated participation entries
6. Filtering relevant variables for analysis

These preprocessing techniques improved the accuracy and readability of the dataset.

3.3. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to identify:

- Participation distribution across countries
- Organizational involvement
- Sector-wise project distribution
- Collaboration trends among institutions

Grouping and aggregation methods were applied to analyse participation frequency and project counts. Visualizations were created to better understand trends and patterns within the data.

4. RESULTS AND FINDINGS

The analysis revealed several important insights regarding European research and innovation projects.

4.1. Horizon 2020 vs Horizon Europe Project Comparison

The comparison between Horizon 2020 and Horizon Europe projects reveals a substantial difference in research activity across the two European framework programmes. As illustrated in Figure 1, Horizon 2020 accounts for approximately 35,389 projects, significantly exceeding the 19,336 projects currently recorded under Horizon Europe. This difference is primarily influenced by the longer operational period and completed lifecycle of Horizon 2020, while Horizon Europe

remains an ongoing programme with continuously expanding participation. The results highlight the large scale and sustained growth of European research and innovation initiatives, demonstrating the European Union's long-term commitment to collaborative scientific development and international research cooperation.

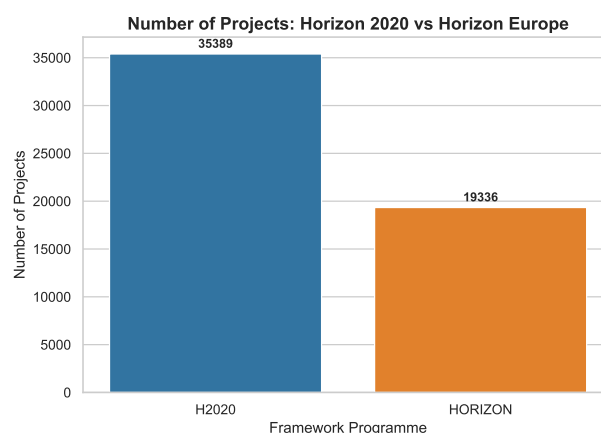


Figure 1. Number of projects in Horizon 2020 and Horizon Europe programmes.

4.2. Framework Programme Budget Analysis

The budget analysis presented in Figure 2 reveals that Horizon 2020 received the highest overall project investment, with total funding exceeding 83 billion, while Horizon Europe accounted for approximately 54 billion in project budgets.

The larger budget allocation under Horizon 2020 is primarily associated with its completed operational cycle and extensive project portfolio. In contrast, Horizon Europe remains an ongoing programme with continuously expanding funding activities.

Overall, the results demonstrate the European Union's sustained financial commitment toward strengthening collaborative research, scientific innovation, and cross-border technological development across successive framework programmes.

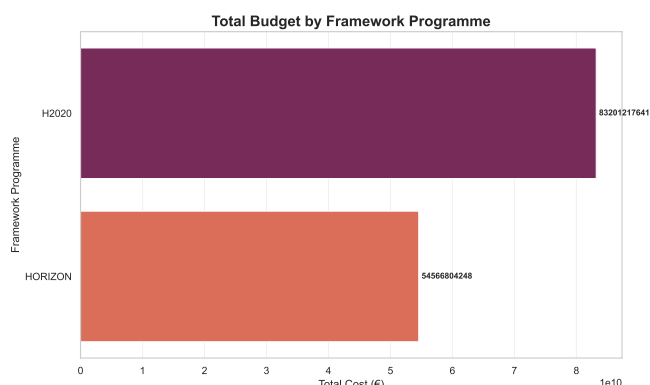


Figure 2. Total budget allocation across European framework programmes.

4.3. Country Participation and Funding Analysis

The analysis revealed that certain European countries (Top 20) had significantly higher participation in Horizon 2020 and Horizon Europe projects. As illustrated in Figure 3, Germany, Spain, Italy, and France emerge as the dominant contributors within the European research collaboration ecosystem, exhibiting the highest levels of project participation, organisational involvement, and funding allocation.

Germany leads across all indicators, with more than 34,000 project participations and over EUR 18 billion in EC funding, highlighting its central role in European innovation networks. Spain and France also demonstrate strong participation and funding performance, while countries such as the Netherlands and Belgium show high funding efficiency relative to their organisational size.

The heatmap analysis further reveals a highly concentrated collaboration structure in which a small number of countries account for a substantial share of European research activity and funding distribution. These findings suggest that European research ecosystems are strongly influenced by a limited group of highly interconnected countries that play a central role in cross-border scientific collaboration and innovation funding.

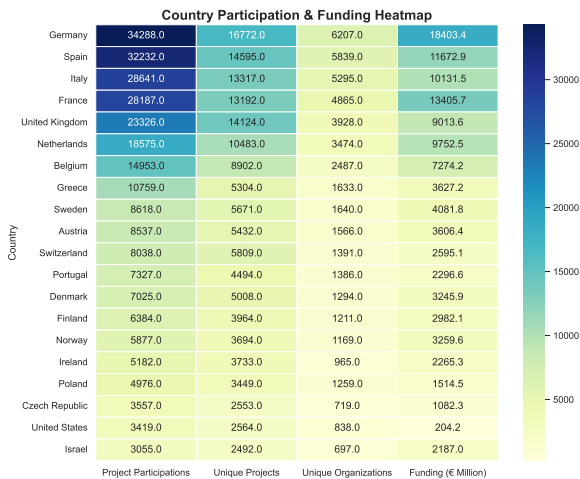


Figure 3. Country participation and funding heatmap.

4.4. European Commission Projects: Duration, Timeline & Budget Insight

The project timeline analysis presented in Figure 4 indicates that most Horizon projects have durations between 2 and 5 years, with average project durations remaining relatively stable across both Horizon 2020 and Horizon Europe programmes.

Project activity increased significantly after 2015, reaching its highest level during the 2019–2023 period, reflecting the expansion of European research and innovation initiatives. Similarly, total project budgets show strong growth over time, peaking in recent years before declining slightly for ongoing or incomplete project cycles.

Overall, the results suggest a consistent long-term investment strategy within European research programmes, characterized by stable project durations, increasing participation, and growing funding intensity across successive framework programmes.

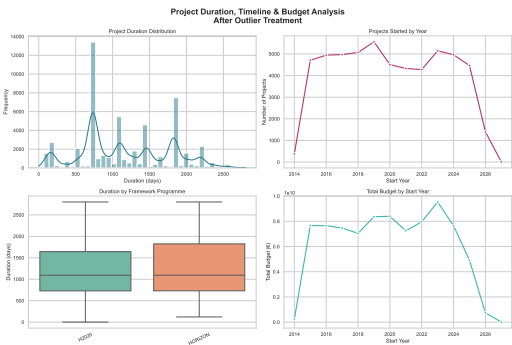


Figure 4. Project duration, timeline, and budget analysis across European framework programmes.

4.5. Integrated Collaboration Network Analysis

The collaboration network results reveal a highly concentrated and interconnected European research ecosystem. As shown in Figure 5, a small number of highly central organisations, including *Fraunhofer*, *CNRS*, *CEA*, and *CSIC*, act as major collaboration hubs connecting universities, research institutes, and industrial partners across Horizon projects.

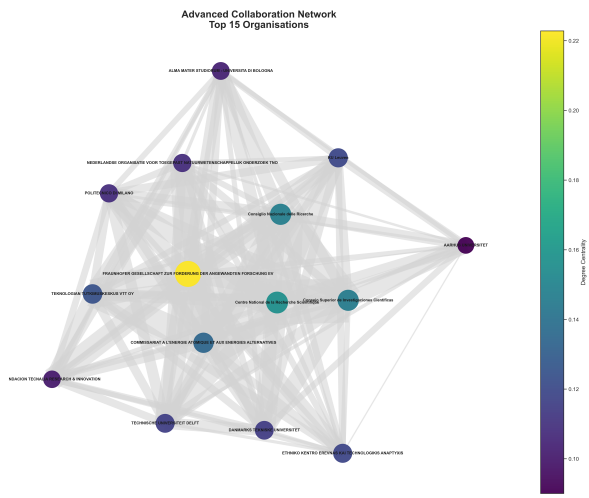


Figure 5. Advanced collaboration network of the top 15 organisations.

At the country level, Figure 6 shows that Germany, Spain, Italy, France, and the United Kingdom form the strongest cross-border collaboration partnerships. This indicates that European research co-operation is strongly shaped by a small group of highly active countries with repeated participation in joint projects.

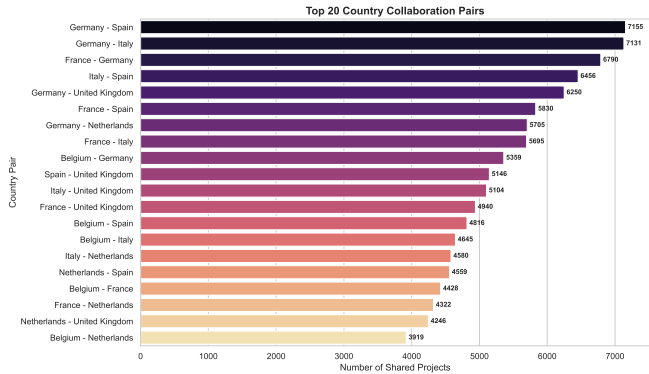


Figure 6. Top 20 country collaboration pairs.



Figure 7. Top 15 strongest organisation collaboration pairs.

Figure 7 further demonstrates that strong institutional partnerships are not random, but often repeated across multiple projects. These long-term collaboration ties suggest the existence of stable strategic alliances within the European research and innovation ecosystem. Overall, the findings show that Horizon collaboration networks are driven by both country-level cooperation and organisation-level hubs

that play a critical role in knowledge exchange and large-scale scientific coordination.

4.6. University–Industry Collaboration Analysis

The university–industry collaboration analysis reveals that a substantial proportion of Horizon projects involve cross-sector partnerships between academic institutions and industrial organisations. As illustrated in Figure 8, approximately 14,043 projects demonstrate direct university–industry collaboration, highlighting the important role of knowledge transfer and applied innovation within European research programmes.

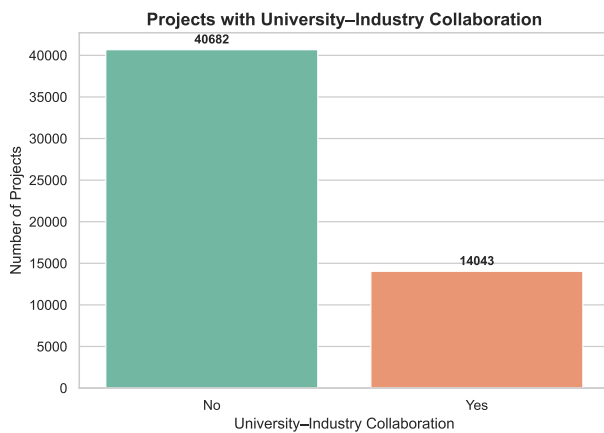


Figure 8. Projects with university–industry collaboration.

However, the majority of projects (40,682) do not include such collaborations, suggesting that many research initiatives remain academically or institutionally concentrated. Overall, the findings indicate that while university–industry partnerships represent a significant component of the European innovation ecosystem, there remains considerable potential to strengthen cross-sector collaboration and industry participation in future research programmes.

4.7. EU Contribution Analysis

The EU contribution analysis highlights a strong concentration of research funding among a limited number of countries and organisations. As illustrated in Figure 10, Germany received the highest overall EU contribution, followed by France, Spain, and Italy, demonstrating their dominant role in European research and innovation activities. At the organisational level, Figure 9 shows that institutions such as

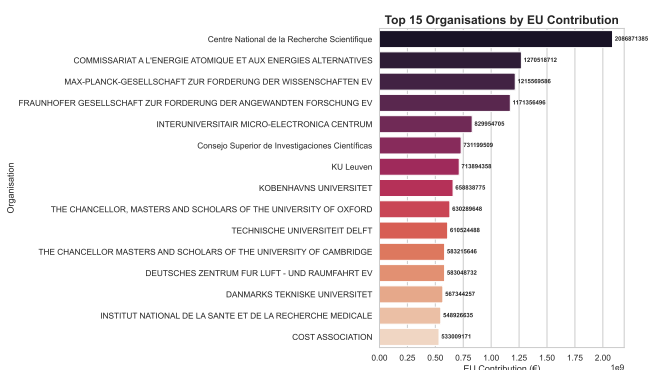


Figure 9. Top 15 organisations by EU contribution.

Centre National de la Recherche Scientifique (CNRS), CEA, Max Planck Society, and Fraunhofer Society emerged as the leading recipients of EU funding, reflecting their central position within Horizon collaboration networks. Overall, the results suggest that European research

funding is highly concentrated among established research-intensive countries and globally recognized scientific institutions.

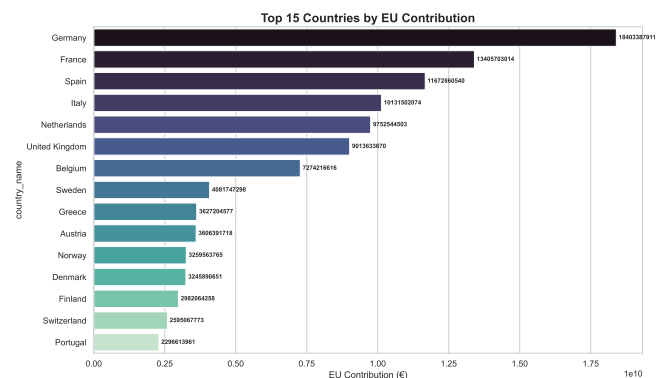


Figure 10. Top 15 countries by EU contribution.

4.8. Project Status and Budget Distribution Analysis

The analysis presented in Figure 11 reveals that most European research projects are either **closed** (56.9%) or currently **signed and ongoing** (39.6%), while only a very small proportion of projects were terminated (3.6%). This indicates a high level of project completion and continuity within the Horizon research programmes.

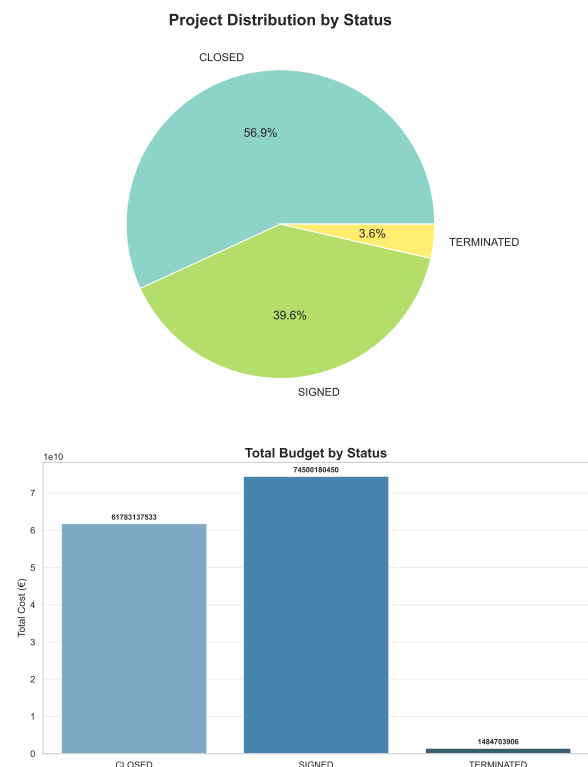


Figure 11. Project status distribution and total budget allocation by project status.

In terms of financial allocation, signed projects account for the largest share of total funding, followed by closed projects, while terminated projects represent only a minor fraction of overall project budgets. Overall, the findings demonstrate the stability and effectiveness of European research funding mechanisms, with strong investment concentrated in active and successfully completed projects.

4.9. Predictive Analysis

Machine learning model was applied to predict whether EU-funded projects belong to *Short*, *Medium*, or *Long* duration categories based on project characteristics such as funding amount, consortium size, funding scheme, and country diversity. The model achieved a very high accuracy of **97.9%**, demonstrating strong predictive performance. The confusion matrix shown in Figure 12 indicates that the model correctly classified the majority of projects across all duration categories with very few misclassifications. Feature importance analysis further revealed that the maximum EU contribution (*ecmaxcontribution*) was the most influential factor affecting project duration, followed by funding scheme and consortium size, as illustrated in Figure 13. The results suggest that highly funded and internationally collaborative projects are more likely to have longer durations, highlighting the usefulness of machine learning techniques in understanding structural patterns within European research and innovation projects.

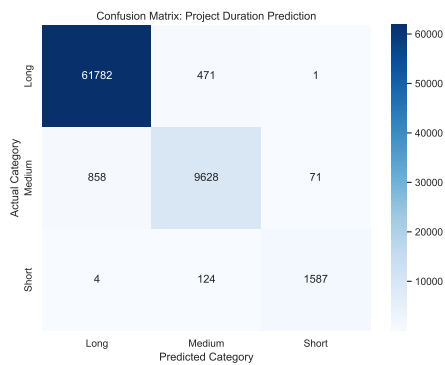


Figure 12. Confusion matrix for project duration prediction.

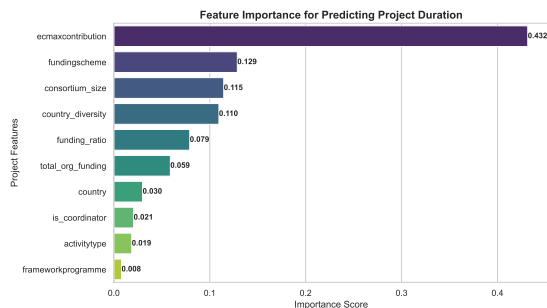


Figure 13. Feature importance for predicting project duration categories.

5. INTERPRETATION AND DISCUSSION

The findings of this study demonstrate that European research and innovation ecosystems are highly collaborative and dominated by a few central countries and organisations, particularly Germany, Spain, France, Italy, and the United Kingdom. Network analysis identified organisations such as *Fraunhofer*, *CNRS*, *CEA*, and *CSIC* as key collaboration hubs that facilitate international research partnerships and knowledge exchange across Europe. Although university–industry collaborations were observed in several projects, the results indicate that cross-sector cooperation remains relatively limited and could be further strengthened in future European funding programmes.

In addition, the predictive analysis using a Random Forest machine learning model achieved an accuracy of **97.9%** in classifying project duration categories (*Short*, *Medium*, and *Long*). The analysis revealed that EU funding contribution, funding scheme, consortium size, and country diversity were the most influential factors affecting project duration. Overall, the study highlights the value of data-driven and machine learning approaches for understanding collaboration

structures, funding dynamics, and strategic research patterns within European innovation ecosystems.

6. CONCLUSION

This project presented a comprehensive exploratory analysis of European research and innovation projects funded under the Horizon 2020 and Horizon Europe programmes using publicly available CORDIS data. The study applied Python-based preprocessing, exploratory data analysis (EDA), and network analysis techniques to investigate project participation, collaboration structures, funding distribution, and organisational involvement across European research ecosystems.

The analysis revealed that European research activities are highly concentrated among a relatively small number of countries and organisations. Germany, France, Spain, Italy, and the United Kingdom emerged as dominant contributors in terms of participation, funding, and collaboration intensity. In addition, major research institutions such as *Fraunhofer*, *CNRS*, and *CEA* were identified as central collaboration hubs within Horizon research networks.

The findings also demonstrated the importance of long-term international partnerships and cross-border scientific cooperation in shaping European innovation ecosystems. While university–industry collaborations were present in a significant number of projects, the results suggest that further integration between academia and industry could strengthen innovation capacity and technology transfer within Europe.

A key contribution of this project was the development of a structured preprocessing pipeline to clean, harmonise, and integrate large-scale CORDIS datasets. The study highlights how data science techniques and visualization methods can provide meaningful insights into complex research collaboration systems and support evidence-based policy analysis.

Future work may include advanced machine learning approaches, predictive modeling, temporal network analysis, and topic modeling techniques to further explore collaboration evolution, funding prediction, and emerging research trends within European research programmes.

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GITHUB REPOSITORY

Visit the GitHub repository to access project data analysis details, visualizations, and report files related to the analysis of Horizon 2020 and Horizon Europe research and innovation datasets.

🔗 https://github.com/nitinmpanwar/sobigdata_project.git

Additional figures and supplementary visualizations can be accessed through the GitHub Repository.